Weather Research and Forecast Model: Avalon Peninsula Weather Simulation

Vahid Sheigani

Memorial University of Newfoundland

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1 Introduction

The Weather Research and Forecasting (WRF) model is a comprehensive meteorological simulation system designed for both operational forecasting and atmospheric research. In this section, we explain the governing mathematical equations of this model, the core components of WRF model and the inner workings of these components. The main source of the WRF model description is the original article, "*A Description of the Advanced Research WRF*", published by NCAR.

1.1 Core Components of WRF

WRF model consists of multiple components that handle, the physics of the model, numerical schemes, data interpolation, WRF chemistry, and different types of simulations. The details of most of these components are beyond the scope of this research. Figure 1 depicts WRF components and the model's workflow. The main components of the WRF model that are related to this research are described below.

- Dynamical Core: The dynamical core of WRF includes the fundamental equations governing fluid motion and thermodynamics in the atmosphere. These are the conservation laws for mass, momentum, and energy, specifically tailored for atmospheric conditions.

- Physical Parameterizations: These are schemes that represent sub-grid scale processes like cloud formation, land surface interactions, radiative transfer, and turbulence. Since these processes occur at scales smaller than the model grid can resolve, they must be approximated or "parameterized."

- Data Assimilation System: WRF includes a data assimilation component which integrates observational data into the model. This improves the initial conditions of forecasts and can be crucial for accuracy.

- Real and Idealized Simulation Frameworks: WRF offers frameworks for both real-world atmospheric conditions and idealized scenarios. This flexibility is key for research applications where specific atmospheric phenomena or theoretical situations need to be studied in isolation.

1.2 ARW Solver

The Advanced Research WRF (ARW) solver is one of the two dynamical solvers available within the Weather Research and Forecasting (WRF) model. The ARW solver uses a set of fully compressible, non-hydrostatic conserved Euler equations in the η coordinate system. These equations account for mass, momentum, and energy conservation. This solver employs a third-order Runge-Kutta scheme, combined with a split-explicit time integration method for the acoustic and gravity-wave modes. For

Workflow

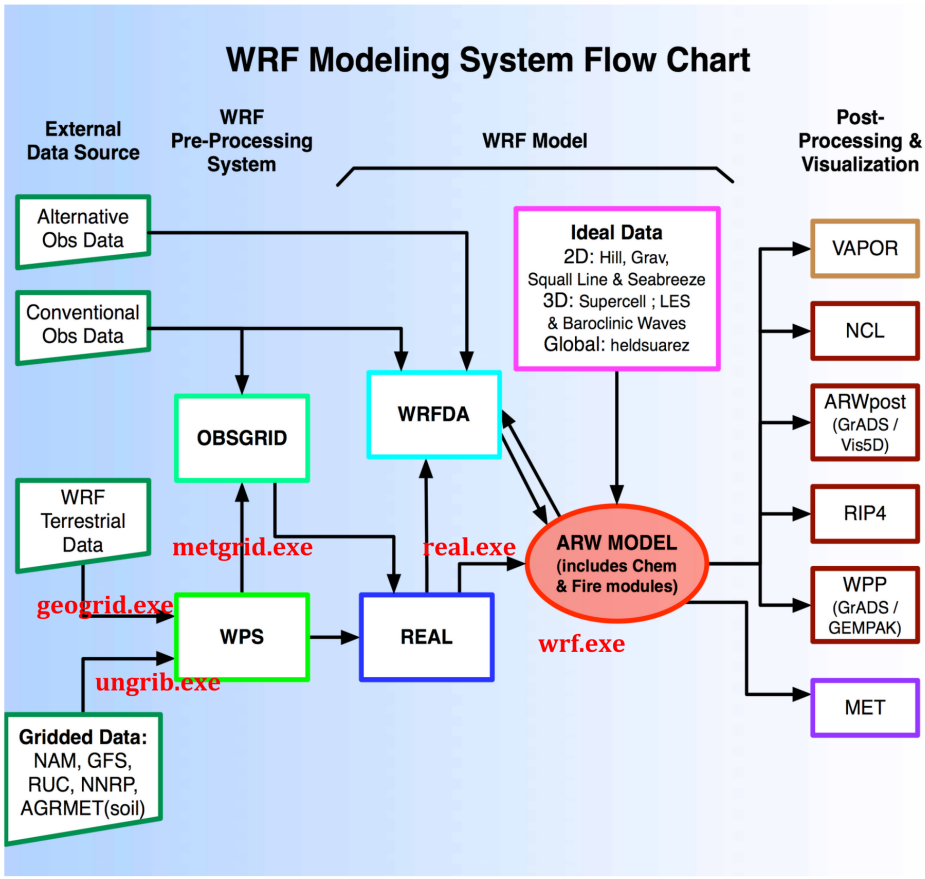

Figure 1: WRF model components, structure and workflow

horizontal discretization, it utilizes a staggered Arakawa C-grid, which helps preserve the symmetries of the flow. The vertical discretization is accomplished using a mass-based terrain-following hydrostatic pressure coordinate system commonly referred to as the η coordinate. Here's an overview of how the ARW solver functions within this coordinate system:

1.2.1 Eta Coordinate System

The η coordinate is a dimensionless vertical coordinate that is scaled to vary from 0 at the top of the model to 1 at the Earth's surface, which allows for a more accurate representation of the atmospheric boundary layer and terrain-induced flows. In this coordinate system, the vertical coordinate is transformed such that the surfaces of constant η follow the terrain at lower levels and become approximately horizontal at

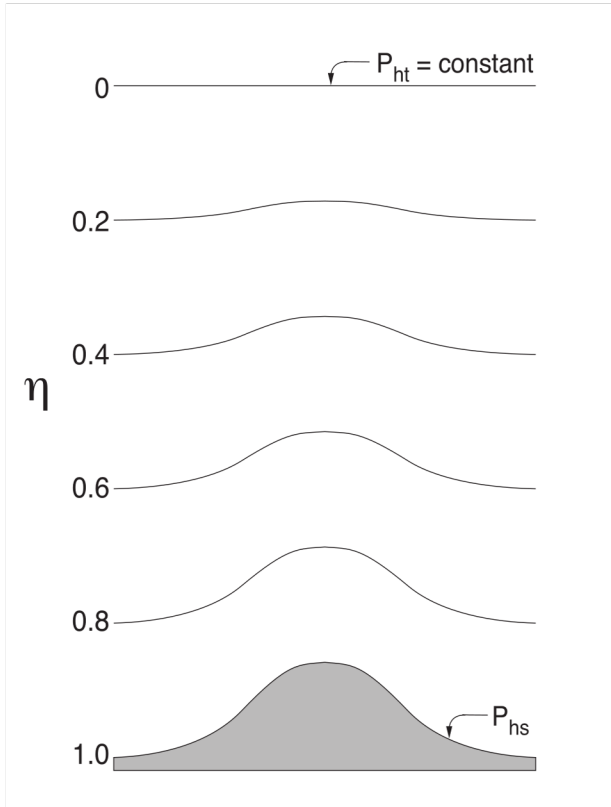

Figure 2: ARW η coordinate system

higher levels. The general form of η coordinate is typically defined as follows

$$\eta = \frac{p - p_{\text{top}}}{p_{\text{surface}} - p_{\text{top}}} \tag{1}$$

where p is the pressure at a given point, p_{surface} is the pressure at the earth's surface, and p_{top} is the pressure at the top of the model domain (usually at the tropopause or lower stratosphere).

1.3 Governing Equations

The equations used in WRF are derived from the fundamental laws of physics applied to atmospheric conditions. Here are the types of equations involved:

1.3.1 Conservation of momentum

These are the Navier-Stokes equations adapted for rotational and stratified flows typical in the atmosphere.

$$\frac{\partial U}{\partial t} + (\nabla \cdot \mathbf{V})u - \frac{\partial}{\partial x}(p\frac{\partial \phi}{\partial \eta}) + \frac{\partial}{\partial \eta}(p\frac{\partial \phi}{\partial x}) = F_U \tag{2}$$

$$\frac{\partial V}{\partial t} + (\nabla \cdot \mathbf{V})v - \frac{\partial}{\partial y}(p\frac{\partial \phi}{\partial \eta}) + \frac{\partial}{\partial \eta}(p\frac{\partial \phi}{\partial y}) = F_V \tag{3}$$

$$\frac{\partial W}{\partial t} + (\nabla \cdot \mathbf{V})w - g(\frac{\partial p}{\partial \eta} - \mu) = F_W \tag{4}$$

where $\mathbf{V} = \mu\mathbf{v} = (U, V, W)$, $\phi = gz$ is geopotential, η is the vertical coordinate, and F_U, F_V, F_W are flux terms in x, y, and z directions.

1.3.2 Conservation of Energy

Represented by the thermodynamic energy equation, accounting for some of the possible methods of heat transfer.

$$\frac{\partial \Theta}{\partial t} + (\nabla \cdot \mathbf{V})\Theta = F_\Theta \tag{5}$$

where $\nabla \cdot \mathbf{v}a = \frac{\partial Ua}{\partial x} + \frac{\partial Va}{\partial y} + \frac{\partial \Omega a}{\partial \eta}$, $\Omega = \mu\eta$, and $\Theta = \mu\theta$. In WRF model the non-conserved form of geo-potential ($\phi = gz$) is used because there is no actual computational benefit to the conserved form. To handle the changes in geo-potential the following equation is used

$$\frac{\partial \phi}{\partial t} + \mu^{-1}[(\mathbf{V} \cdot \nabla)\phi] - gW = 0 \tag{6}$$

1.3.3 Conservation of Mass

Often represented by the continuity equation for incompressible flow, modified for compressible atmospheric conditions.

$$\frac{\partial \mu}{\partial t} + (\nabla \cdot \mathbf{V}) = 0 \tag{7}$$

where $\mu(x, y)$ is mass per unit area within the columns, and $(\mathbf{V})$ is the velocity vector.

1.3.4 Conservation of Water-vapor

Conservation of water-vapor in atmospheric dynamics is crucial in meteorological modelling for predicting humidity, cloud formation, precipitation, and related weather phenomena. To adjust our equations for water-vapour conservation, we need to switch μ with μ_d in equations 6 and 7. We also need to make sure that the total mass of water-vapor is conserved. Equation 8 ensures the inclusion of phase changes of water (evaporation, condensation, freezing, and melting) and their latent heat effects in the WRF model.

$$\frac{\partial Q_m}{\partial t} + (\nabla \cdot \mathbf{V}_{q_m}) = F_{Q_m} \tag{8}$$

where μ_d is the mass per unit area of dry-air, q_m is the mixing ratio (mass per mass of dry air) for water vapor, and $Q_m = \mu_d q_m$.

1.4 Map Projections

1.4.1 Physical Distances Between Grid Points

Both isotropic and anisotropic transformations are included in WRF version 3. Projections to the sphere require that the physical distances between grid points in the projection vary with position on the grid. This addresses the fact that the Earth is a sphere, and representing it on a flat grid (like in a model) requires adjusting for this curvature.

- Isotropic transformations require that the ratio $(\Delta x/\Delta y)_{\text{earth}}$ be constant everywhere on the grid. This means the grid spacing in the x-direction to the grid spacing in the y-direction remains constant, preserving angles and shapes locally.

$$\left(\frac{\Delta x}{\Delta y}\right) = \text{constant everywhere on the grid} \tag{9}$$

- Anisotropic transformations need the definition of map factors. These are used to correct the grid distances on a map projection to their actual distances on the Earth's surface. The map factors are represented as (m_x, m_y), and they relate the model's grid spacing $(\Delta x, \Delta y)$ to the physical distances on the Earth.

$$(m_x, m_y) = \frac{(\Delta x, \Delta y)}{\text{distance on the earth}} \tag{10}$$

1.4.2 Map-Scale Factors in Governing Equations

The ARW solver includes map-scale factors in the governing equations by redefining the momentum variables as follows:

$$U = \frac{\mu_d u}{m_y}, \quad V = \frac{\mu_d v}{m_x}, \quad \frac{\mu_d w}{m_y}, \quad \Omega = \frac{\dot{\eta}}{m_y} \tag{11}$$

These equations and transformations are vital for accurate numerical weather prediction since they ensure that the model's calculations correctly represent the actual distances and areas over the Earth's curved surface.

1.5 Coriolis and Curvature

The right-hand side of the equations of the conservation of momentum in U, V, and W directions (equations 2, 3, and 4) include terms that account for the Coriolis effect and curvature of the earth.

1. **Isotropic Projections**: For isotropic projections such as Lambert conformal, polar stereographic, and Mercator, where $(m_x = m_y = m)$, the Coriolis and curvature terms are defined in the following way:

$$F_{U_{cor}} = +\left(f + u\frac{\partial m}{\partial y} - v\frac{\partial m}{\partial x}\right)V - eW\cos\alpha_r - \frac{uW}{r_e} \tag{12}$$

$$F_{V_{cor}} = -\left(f + u\frac{\partial m}{\partial y} - v\frac{\partial m}{\partial x}\right)U + eW\sin\alpha_r - \frac{vW}{r_e} \tag{13}$$

$$F_{W_{cor}} = +e\left(U\cos\alpha_r - V\sin\alpha_r\right) + \left(\frac{uU + vV}{r_e}\right) \tag{14}$$

where α_r is the local rotation angle between the y-axis and the meridians, Ω_e is the Earth's angular rotation rate, $f = 2\Omega_e \sin\psi$ is the Coriolis parameter, $e = 2\Omega_e \cos\psi$, r_e is the radius of the earth, and ψ in the definitions of f and e is the latitude.

2. **Anisotropic Projections**: The curvature and Coriolis terms for the momentum equations are cast in the following form for the anisotropic latitude-longitude grid:

$$F_{U_{cor}} = \frac{m_x}{m_y}\left[fV + \frac{uV\tan\phi}{r_e}\right] - \frac{uW}{r_e} - eW\cos\alpha_r \tag{15}$$

$$F_{V_{cor}} = \frac{m_y}{m_x}\left[-fU - \frac{uU\tan\phi}{r_e}\right] - \frac{vW}{r_e} + eW\sin\alpha_r \tag{16}$$

$$F_{W_{cor}} = +e(U\cos\alpha_r - (m_x/m_y)V\sin\alpha_r) + \left(\frac{uU + (m_x/m_y)vV}{r_e}\right) \tag{17}$$

where α_r is the local rotation angle between the y-axis and the meridians, Ω_e is the Earth's angular rotation rate, $f = 2\Omega_e \sin\psi$ is the Coriolis parameter, $e = 2\Omega_e \cos\psi$, r_e is the radius of the earth, and ψ in the definitions of f and e is the latitude.

1.5.1 Final Governing equations

Using these redefined momentum variables, the governing equations, including map factors and rotational terms, can be written as

$$\frac{\partial U}{\partial t} + m_x \left[\frac{\partial}{\partial x}(Uu) + \frac{\partial}{\partial y}(Vu) \right] + \frac{\partial}{\partial \eta}(\Omega u) + \mu_d \alpha \frac{\partial}{\partial x} p + (\alpha/\alpha_d) \frac{\partial}{\partial \eta} p \frac{\partial \phi}{\partial x} = F_U \tag{18}$$

$$\frac{\partial V}{\partial t} + m_y \left[\frac{\partial}{\partial x}(Uv) + \frac{\partial}{\partial y}(Vv) \right] + (m_y/m_x) \frac{\partial}{\partial \eta}(\Omega v) + \mu_d \alpha \frac{\partial}{\partial y} p + (\alpha/\alpha_d) \frac{\partial}{\partial \eta} p \frac{\partial \phi}{\partial y} = F_V \tag{19}$$

$$\frac{\partial W}{\partial t} + (m_x m_y/m_y) \left[\frac{\partial}{\partial x}(Uw) \right] + \frac{\partial}{\partial y}(Vw)] + \frac{\partial}{\partial \eta}(\Omega w) - m_y^{-1} g (\alpha/\alpha_d) \frac{\partial}{\partial \eta} p - \mu_d = F_W \tag{20}$$

$$\frac{\partial \Theta}{\partial t} + m_x m_y \left[\frac{\partial}{\partial x}(U\Theta) + \frac{\partial}{\partial y}(V\Theta) \right] + m_y \frac{\partial}{\partial \eta}(\Omega \Theta) = F_\Theta \tag{21}$$

$$\frac{\partial \mu_d}{\partial t} + m_x m_y \left[U_x + V_y \right] + m_y \frac{\partial}{\partial \eta}(\Omega) = 0 \tag{22}$$

$$\frac{\partial \phi}{\partial t} + \mu_d^{-1} \left[m_x m_y (U \frac{\partial}{\partial x} \phi) + V \frac{\partial}{\partial y} \phi) + m_y \Omega \frac{\partial}{\partial \eta} \phi - m_y g W \right] = 0 \tag{23}$$

$$\frac{\partial Q_m}{\partial t} + m_x m_y \left[\frac{\partial}{\partial x}(UQ_m) + \frac{\partial}{\partial y}(VQ_m) \right] + m_y \frac{\partial}{\partial \eta}(\Omega Q_m) = F_{Q_m} \tag{24}$$

1.5.2 Other Features

- Perturbation form of the governing equations: recasts the equations using perturbation variables to reduce truncation errors in the horizontal pressure gradient calculations in the discrete solver and machine rounding errors in the vertical pressure gradient and buoyancy calculations.

- Microphysics: microphysics schemes ranging from simplified physics suitable for idealized studies to sophisticated mixed-phase physics suitable for process studies and NWP are included in WRF version 3.

- Surface physics: multi-layer land surface models ranging from a simple thermal model to full vegetation and soil moisture models, including snow cover and sea ice.

- Planetary boundary layer physics: turbulent kinetic energy prediction or non-local K schemes.

- Atmospheric radiation physics: long-wave and shortwave schemes with multiple spectral bands and a simple shortwave scheme suitable for climate and weather applications. Cloud effects and surface fluxes are included.

- Turbulent mixing and model filters: grid-nesting, explicit spatial diffusion, vertical velocity damping.

1.6 Numerical Solving Schemes

WRF employs the third-order Runge-Kutta scheme for temporal discretization, the Arakawa C-grid staggering for horizontal discretization, and the terrain-following hydrostatic pressure coordinate for vertical discretization.

1.6.1 Temporal Discretization and Runge-Kutta Time Integration

Low-frequency (meteorologically significant) modes are integrated using a third-order Runge-Kutta (RK3) time integration scheme [5]. High-frequency acoustic modes are integrated over smaller time steps to maintain numerical stability. The horizontally propagating acoustic modes and gravity waves are integrated using a forward-backward time integration scheme. Vertically propagating acoustic modes and buoyancy oscillations are integrated using a vertically implicit scheme (using the acoustic time step). The third-order Runge-Kutta time integration scheme can be represented by the following equations

$$\begin{aligned} \Phi^{*} &= \Phi^{t} + \frac{\Delta t}{3} R(\Phi^{t}) \\ \Phi^{**} &= \Phi^{t} + \frac{\Delta t}{2} R(\Phi^{*}) \\ \Phi^{t+\Delta t} &= \Phi^{t} + \Delta t R(\Phi^{**}) \end{aligned} \tag{25}$$

1.6.2 Spatial Discretization

ARW uses C-grid staggering for the variables. The scalar quantities such as temperature, pressure, or density (thermodynamic variables) are defined at the center of grid cells. The vector quantities like velocity components are defined at the edges of grid cells. This means that the east-west component of the velocity (u) is defined at the center of the east and west faces of the grid cell. Similarly, the north-south component of the velocity (v) is defined at the center of the north and south faces. Figure 3 depicts the shape of the staggered C-grid and the location of all variables.

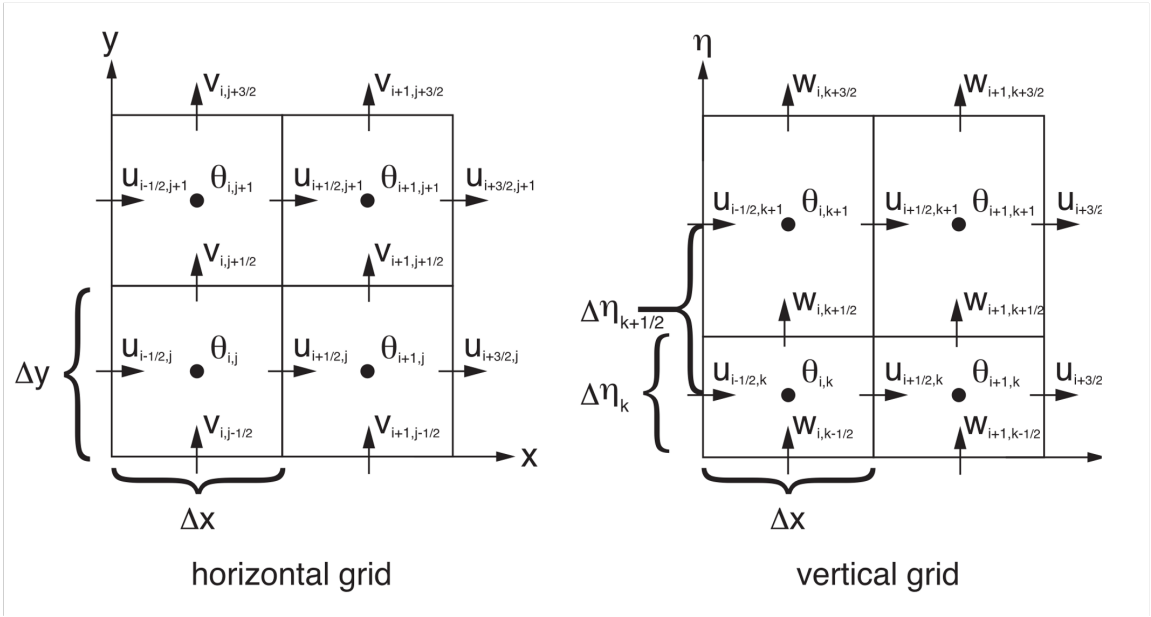

Figure 3: Horizontal and Vertical grids of staggered

2 Method

2.1 Installation Process

Compiling and running the WRF model on macOS requires a similar approach to Linux but with some macOS-specific adjustments, particularly in installing the necessary compilers and libraries. Here's how to adapt each step for macOS:

2.1.1 Environment Setup

macOS does not come with a package manager like apt or yum, but you can use Homebrew, a popular package manager for macOS. First, install homebrew if you haven't already:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/
install/HEAD/install.sh)"
```

To compile WRF, WPS and their prerequisites, `gfortran`, `gcc`, and `cpp` compilers are required. These compilers are then used to compile and install the following required libraries:

```
mpich-3.0.4
netcdf-c-4.7.2
netcdf-fortran-4.5.2
Jasper-1.900.1
libpng-1.2.50
zlib-1.2.11
```

To start the compilation process, first, install gfortran and gcc using `apt` in Ubuntu or `homebrew` in macOS. The following commands can be used in macOS:

```
brew install gcc
brew install gfortran
```

Then download and install all of the required libraries. In most cases, you need the following commands to compile and install these libraries.

```
./configure
make
make install
```

2.1.2 WRF Configuration the Compilation options

Before running the configure script, it's essential to ensure that your environment is set up to use the correct compilers. For macOS, you often need to explicitly set environment variables to use the versions of GCC and gfortran installed by Homebrew:

```
export FC=/usr/local/bin/gfortran
export CC=/usr/local/bin/gcc
./configure
```

2.1.3 Compiling the Model and Preprocessing Software

The compilation commands are almost the same for Ubuntu and macOS. The following sections explain the process of compiling WRF and WPS.

2.1.4 Compiling WRF

```
./configure
./compile em_real >& compile.log
```

2.1.5 Compiling WPS

The following commands should be executed to compile and install WPS:

```
export DIR=/path_to_directory/Build_WRF/LIBRARIES
cd WPS
./clean
export JASPERLIB=$DIR/grib2/lib
export JASPERINC=$DIR/grib2/include
./configure
export WRF_DIR=../WRF
./compile >& log.compile
```

If the compilation is successful, there should be 3 executables in the WPS top-level directory, that are linked to their corresponding `src/` directories:

```
geogrid.exe -> geogrid/src/geogrid.exe
ungrib.exe -> ungrib/src/ungrib.exe
metgrid.exe -> metgrid/src/metgrid.exe
```

If one of the files is missing after the first compilation is completed without any errors, running `./compile >& log.compile` command without rerunning the previous commands will solve the issue. You can also look inside the log files to find the reported error messages by running `less compile.log`.

2.2 Preprocessing

The first step required for running WRF is data-preprocessing. WRF package comes with WPS software that is responsible for data preprocessing and is executed in three steps. Before starting to run the WPS command, we need to modify the `namelist.wps`

file to specify the location and time range of our simulation. I used the following settings in my `namelist.wps`. Setting the geographical location of the model to Avalon Penisula was not easy in the beginning because the manual specification of the latitude and longitude of any region is not an easy task. For this task, I used `WRFDomainWizard` which can be found at https://jiririchter.github.io/WRFDomainWizard/. Using this website you can choose the domain graphically and generate the `&geogrid` section of the following `namelist.wps`. The main source that I used for this part of the research is a tutorial by the University of Waterloo [4]. For further information regarding `namelist.wps` file configuration options and best practices go to https://www2.mmm.ucar.edu/wrf/users/namelist_best_prac_wps.html

```
&share
 wrf_core = 'ARW',
 max_dom = 2,
 start_date = '2019-09-04_12:00:00','2019-09-04_12:00:00',
 end_date   = '2019-09-06_12:00:00','2019-09-06_12:00:00',
 interval_seconds = 10800
 io_form_geogrid      = 2
 debug_level          = 0
/
&geogrid
 parent_id            = 1, 1
 parent_grid_ratio    = 1, 3
 i_parent_start       = 1, 6
 j_parent_start       = 1, 6
 e_we                 = 57, 139
 e_sn                 = 52, 124
 geog_data_res        = 'default', 'default'
 dx                   = 12000
 dy                   = 12000
 map_proj             = 'polar'
 ref_lat              = 47.572
 ref_lon              = -52.734
 truelat1             = 47.572
 pole_lat             = 90
 pole_lon             = 0
 stand_lon            = -52.734
 geog_data_path       = '/Users/Vahid/WRF/WPS/WPS_GEOG_DATA/WPS_GEOG/'
 opt_geogrid_tbl_path = './geogrid/'
/
&ungrib
 out_format = 'WPS',
 prefix = 'GFS'
```

```
/
&metgrid
 fg_name              = 'GFS'
 io_form_metgrid      = 2
 opt_metgrid_tbl_path = './metgrid'
/
```

The complete list of available options for `namelist.wps` file can be found on UCAR website (`https://www2.mmm.ucar.edu/wrf/users/namelist_best_prac_wps.html`)

2.2.1 Geogrid

To run WRF using real data, we need to obtain both static geography data and GRIB (General Regularly-distributed Information in Binary form) data that describes the weather conditions. To be able to execute geogrid command we need to download our geogrid data from the official NCAR website. The required steps are as below: **Geography Data**: For this step, create and change your directory to `/geog` directory. Then download and unzip the high-resolution geography data found in the following link, and execute the `./geogrid` command.

```
wget http://www2.mmm.ucar.edu/wrf/src/wps_files/geog_complete.tar.gz
./geogrid >& log.geogrid
```

Even in WRF V4.0, you will need to use the geographical data provided with WRF V3. Due to an error (as of Oct 24 2018) WRF will ask for data that is only provided in the previous dataset. We are required to put the data path in the appropriate path of our geog data in `namelist.wps` file.

2.2.2 Ungrib

Observations data: In this step we need weather observations data. The reanalysis weather data that I have used for our simulation can be found at the following link. https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html. I only used the September 2019 reanalysis data as the data files take a long time to download. To untar and unzip all the files the following command is used:

```
for a in `ls -1 *.tar`; do tar -xvf $a; done
```

Then we have to link the data files to the WPS directory. To do that we navigate to `/WPS` directory and execute the following command:

```
./link_grib.csh path_to_gfs_datafiles/*\
ln -sf ungrib/Variable_Tables/Vtable.GFS Vtable
./ungrib >& log.ungrid
```

2.2.3 Metgrid

For this step we only need to run the following commands:

```
./metgrid.exe >& log.metgrid
```

2.3 Running WRF

To run WRF, we need to navigate to `/WRF/run` or `WRF/test/em_real` directory. Create a backup of that folder and run the following commands:

```
ln -sf path_to_met_em_files/met_em.d0* .
./real.exe >& log.real
./wrf.exe >& log.wrf
```

A great resource for troubleshooting this process is a PDF file created by NCAR. It can be downloaded from the following URL: https://homepages.see.leeds.ac.uk/~lecrrb/wrf/aRWUsersGuide.pdf.

2.4 Postprocessing

2.4.1 Concatenating of WRF Output Files

To combine WRF output files use the following command

```
ncrcat wrfout_d02_*.nc wrfout_d02_combined.nc
```

2.4.2 Extracting Variable Names and Codes

The easiest method to get the variable names and corresponding codes is installing `cdo` and extracting variable names and codes using the following commands

```
brew install cdo
cdo vardes wrfout_all_combined.nc
```

Choose variables that you want to analyze based on the variable name or variable codes in the output of the above commands and extract them using one of the commands below

```
cdo selvar,variable_name input.nc output.nc
cdo selcode,variable_code input.nc output.nc
```

In this research, I used the following command to extract the variables of my choosing

```
cdo selcode,-1,-4,-5,-6,-12,-13,-20,-21,-24 wrfout_all_combined.nc \
wrfout_sel.nc
```

The variables that are extracted by executing this command are

```
-1  LU_INDEX
-2  ZS
-3  DZS
-5  PHB
-6  T
-7  TSK_FORCE_TEND
-8  MU
-9  P
```

The method explained above has one caveat: It will fail to extract time, latitude, and longitude if there is any inconsistency between any of the variables in the output files. I observed that sometimes a variable definition or boundaries slightly change between different output files and as a result `cdo` command fails. The reason is that `cdo` does not support changing variables. The solution is to use `nc` toolkit to merge and select variables. To do that, we need to extract the variables of all WRF output files first. Running the following shell script will accomplish that.

```bash
#!/bin/bash

file_ls=$(ls wrfout_d02_*)
for file in $file_ls; do
    echo $file
    ncks -v XLONG,XLAT,LANDMASK,XLAND,T2,QVAPOR,PSFC,U10 \
    ,V10,SST,LU_INDEX,PHB,TSK_FORCE_TEND,P,MU "$file" \
    "${file}_sel02".nc
done
```

Then, we can safely merge the output files that include selected variables using the following command

```
ncrcat wrfout_d02_*_sel02.nc wrfout_d02_selcomb02.nc
```

The output of this command is `wrfout_d02_selcomb02.nc` file which is ready for analysis and generating graphs.

3 Results

In this section, we analyze and compare the graphs that are generated based on WRF output files. This simulation is conducted for 48 hours starting at 12 AM on September 4, 2019, and ending at 12 AM on September 6, 2019. There are two types of graphs in this section. The first type of graphs includes the graphs that are generated at the beginning and at the end of the simulation, and the second type of graphs are the time series plots that depict the change of the selected variables over time. The variable names that are chosen for this analysis are U10, V10, T2, SST, PSFC, P, and QVAPOR that refer to u, v, surface temperature, sea surface temperature, surface pressure, pressure at different heights, and water vapor mixing ratio respectively. The graphs that depict the state of a variable at a point in time are generated using the Python code snippet provided in the appendix section, and the time series graphs are generated using Panoply geo-referenced data plotting software developed by NASA. To get more information on this software visit https://www.giss.nasa.gov/tools/panoply/.

3.1 Colorbar and Contour Plots

3.1.1 Surface Temperature

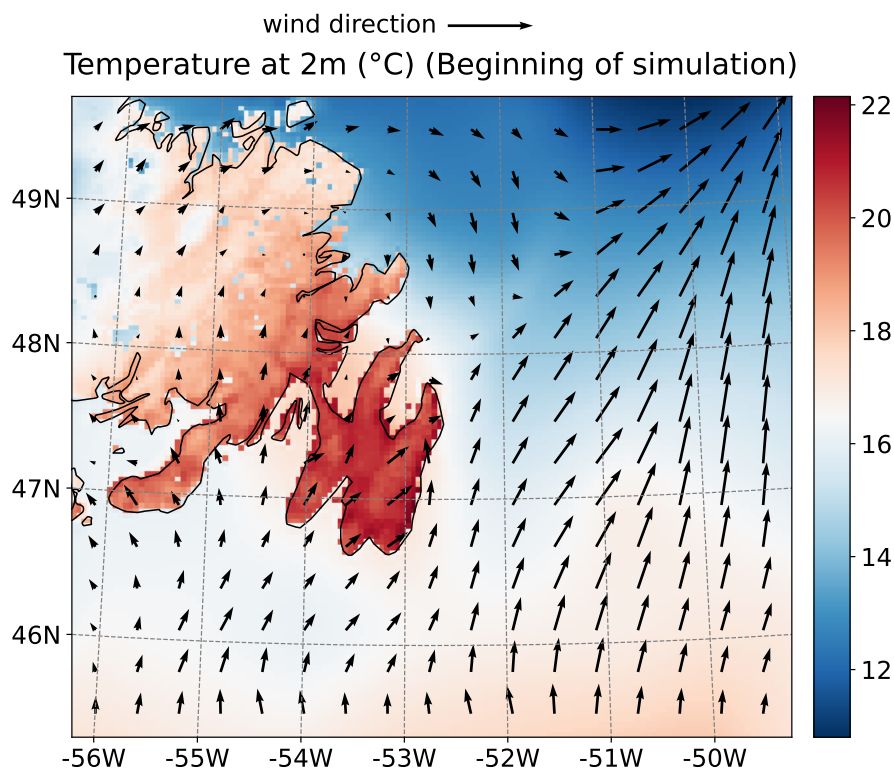

Figure 4: Surface Temperature on Sep 4, 2019

Figures 4 and 5 show surface temperature for the beginning and the end of the simulation respectively. Comparing these graphs conveys an overall decrease in temperature throughout the simulation, and transferring heat from land to sea part of the figure. The arrows on this graph show the relative magnitude and the direction of wind.

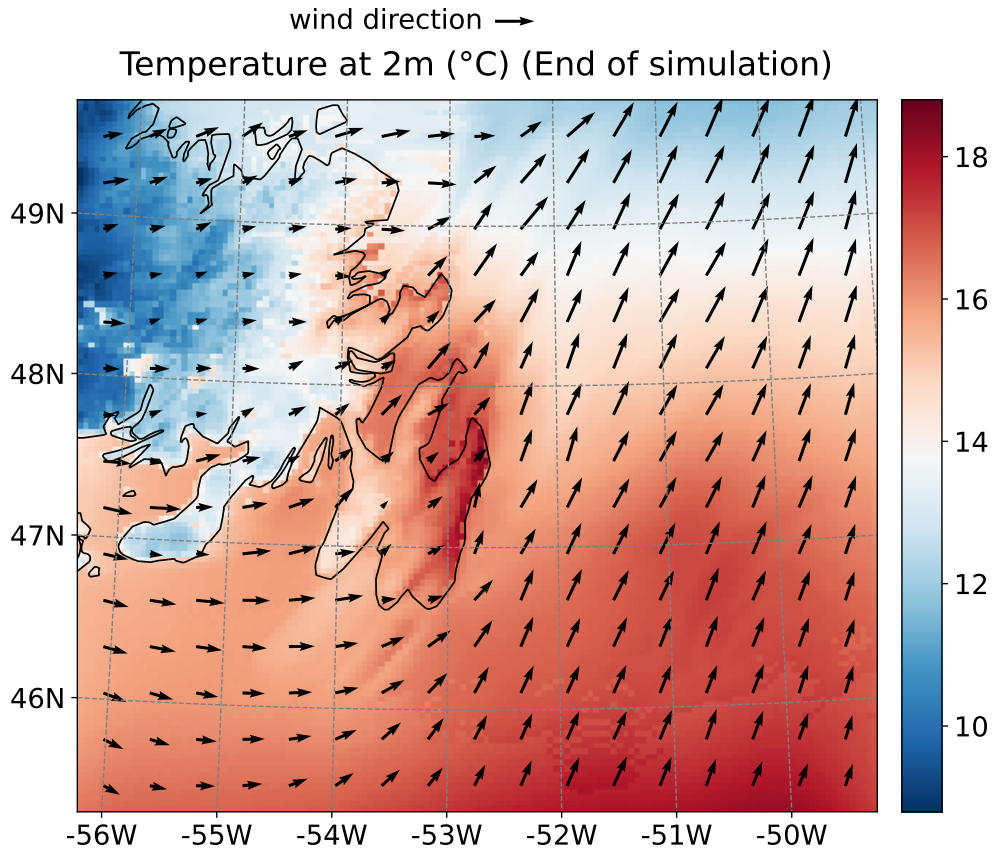

Figure 5: Surface Temperature on Sep 6, 2019

3.1.2 Sea Surface Temperature

Figures 10 and 11 depict sea surface temperature for the beginning and the end of the simulation respectively. Comparing these graphs conveys little to no change in sea surface temperature throughout the simulation. This result agrees with our expectations considering the short length of this simulation.

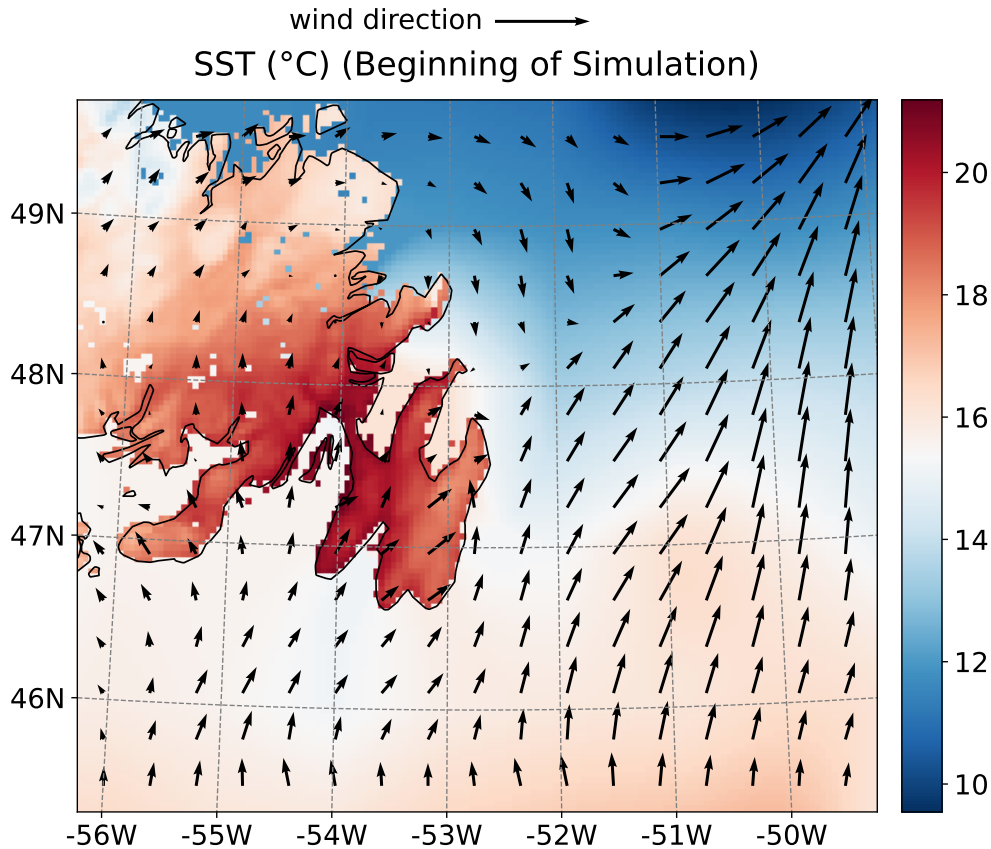

Figure 6: Sea Surface Temperature on Sep 4, 2019

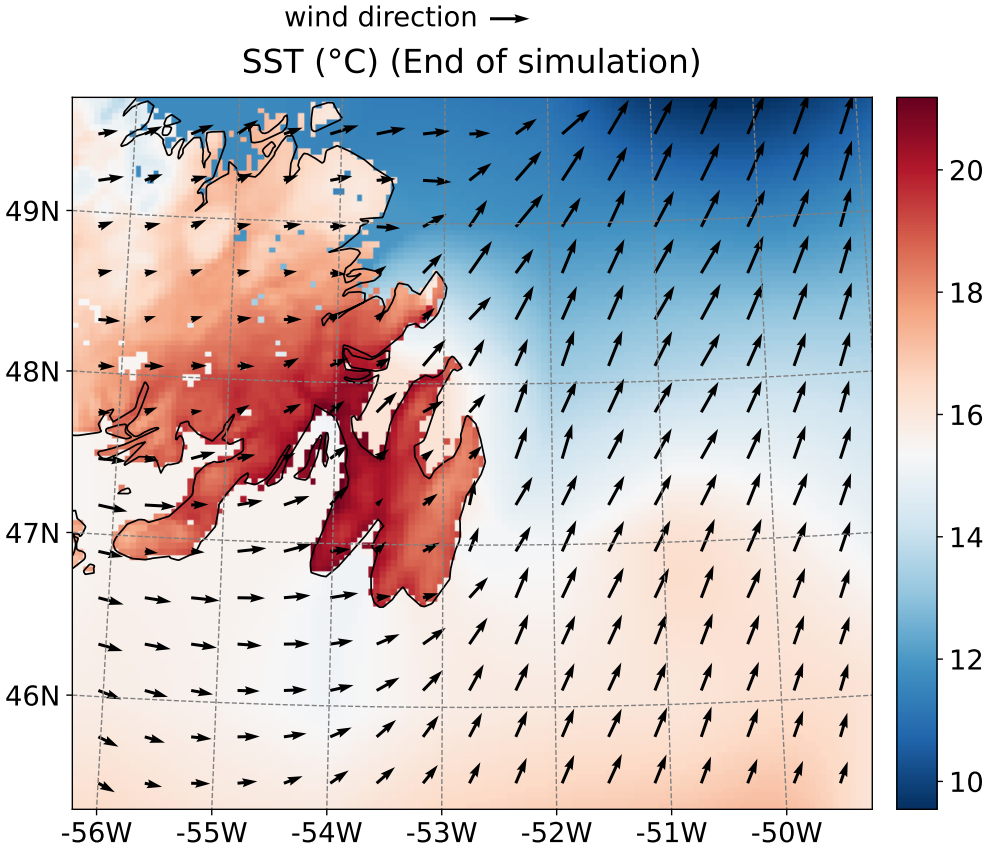

Figure 7: Sea Surface Temperature on Sep 6, 2019

3.1.3 Surface Pressure

Figures 8 and 9 show the surface pressure for the beginning and the end of the simulation respectively. Comparing these graphs conveys a significant overall decrease in pressure throughout the simulation. The graphs confirm that this decrease in pressure has a higher magnitude over the land area. The arrows on this graph show the magnitude and direction of increasing pressure.

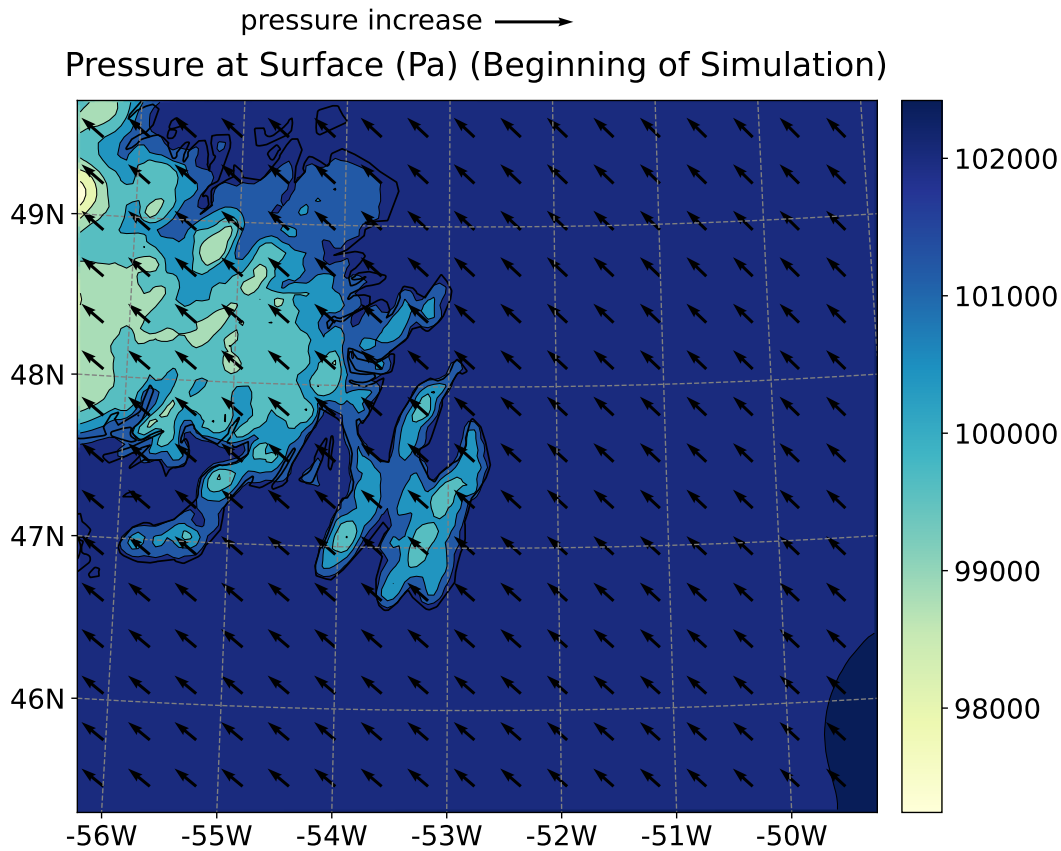

Figure 8: Surface Pressure on Sep 4, 2019

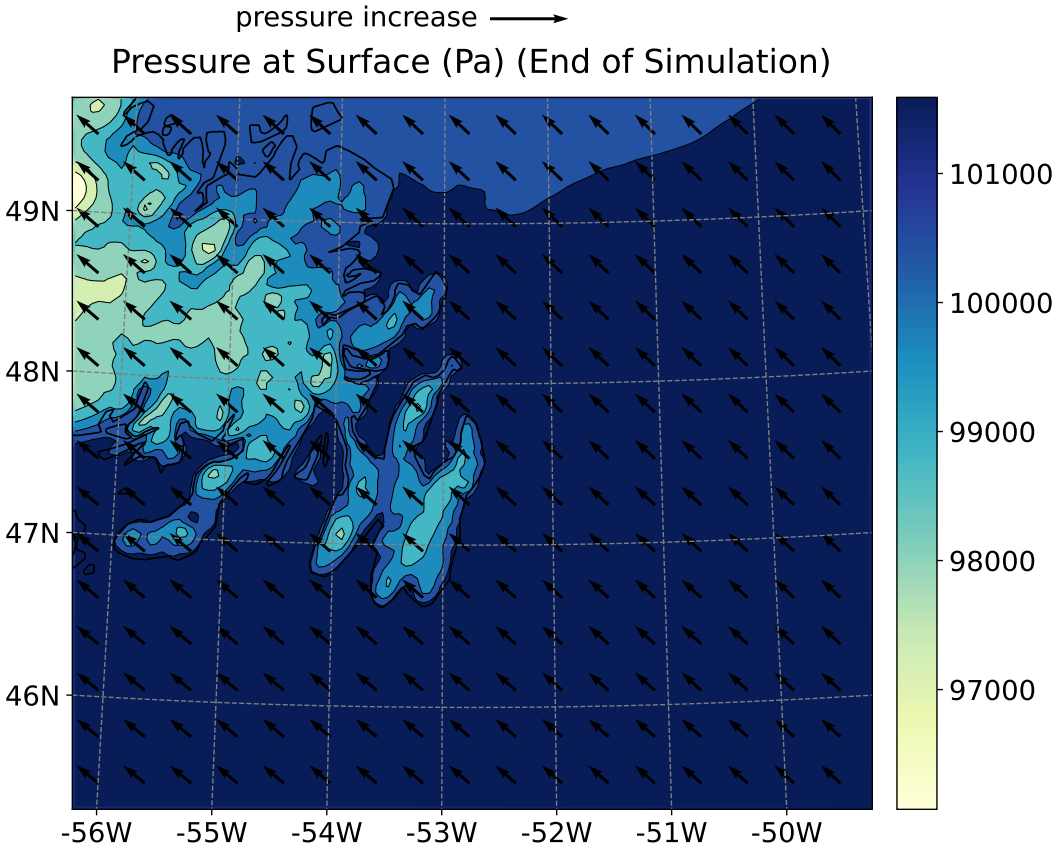

Figure 9: Surface Pressure on Sep 6, 2019

3.1.4 Mid-height Atmosphere Pressure

Figures 10 and 11 show the pressure at approximately 300 meters from the surface for the beginning and the end of the simulation respectively. Comparing these graphs shows a significant overall decrease in pressure especially on the land, and a changing pattern of wind direction throughout the simulation. In addition, at the beginning and end of the simulation pressure above the land area is lower than the pressure above the sea surface.

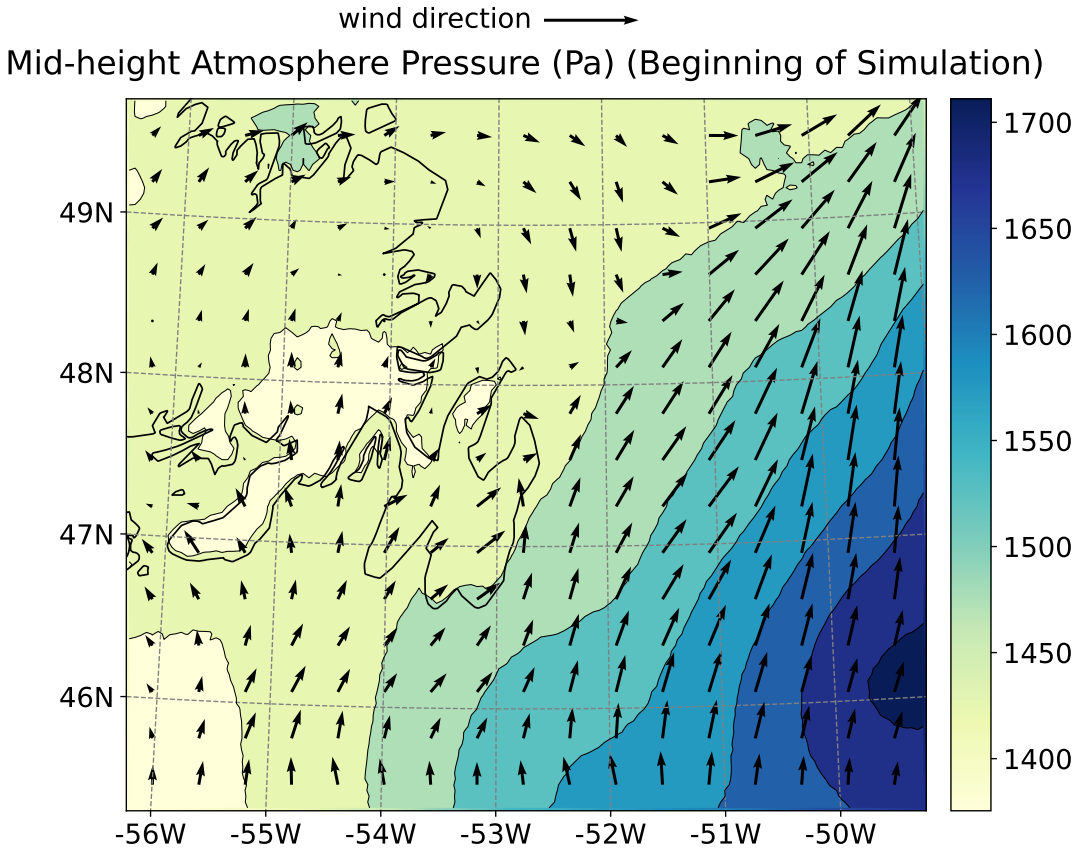

Figure 10: Mid-height Pressure on Sep 4, 2019

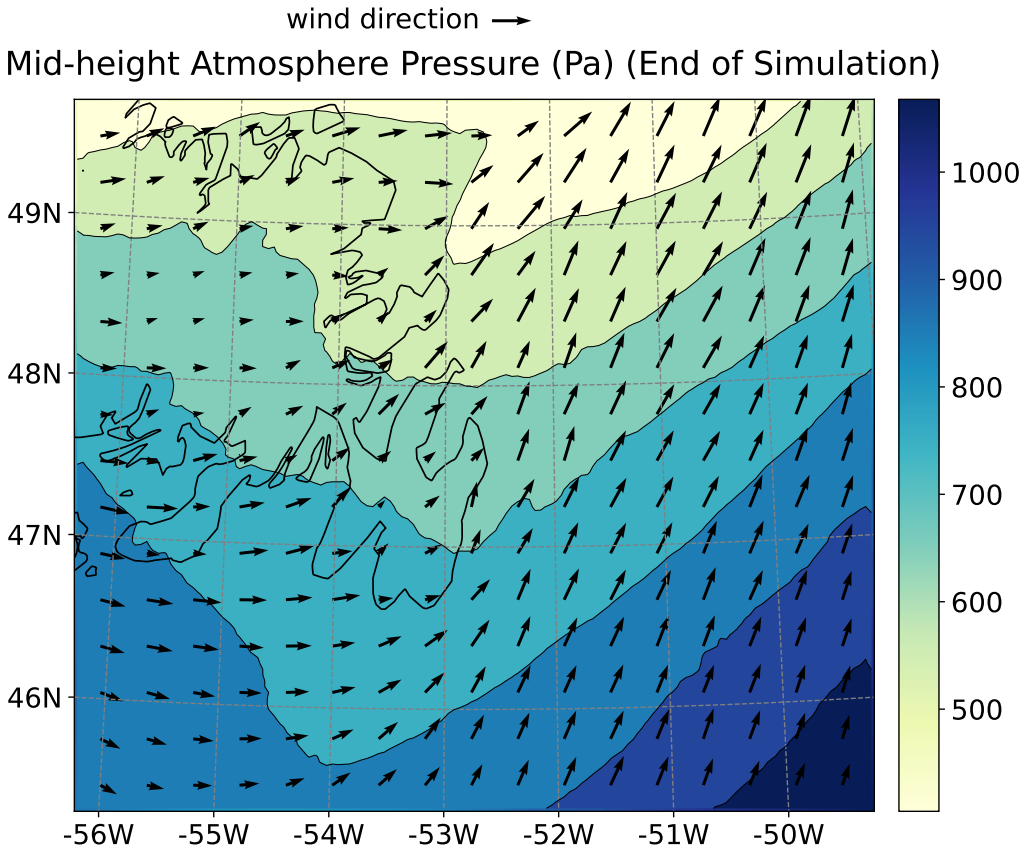

Figure 11: Mid-height Pressure on Sep 6, 2019

3.1.5 Water-vapor Mixing Ratio

Figures 12 and 13 depict the water vapor mixing ratio near the surface at the beginning and the end of the simulation respectively. Comparing these graphs conveys an overall increase of the mixing ratio over the sea surface, and an overall decrease of the mixing ratio over the land, The arrows on this graph show the magnitude and the direction of the decreasing water vapor mixing ratio.

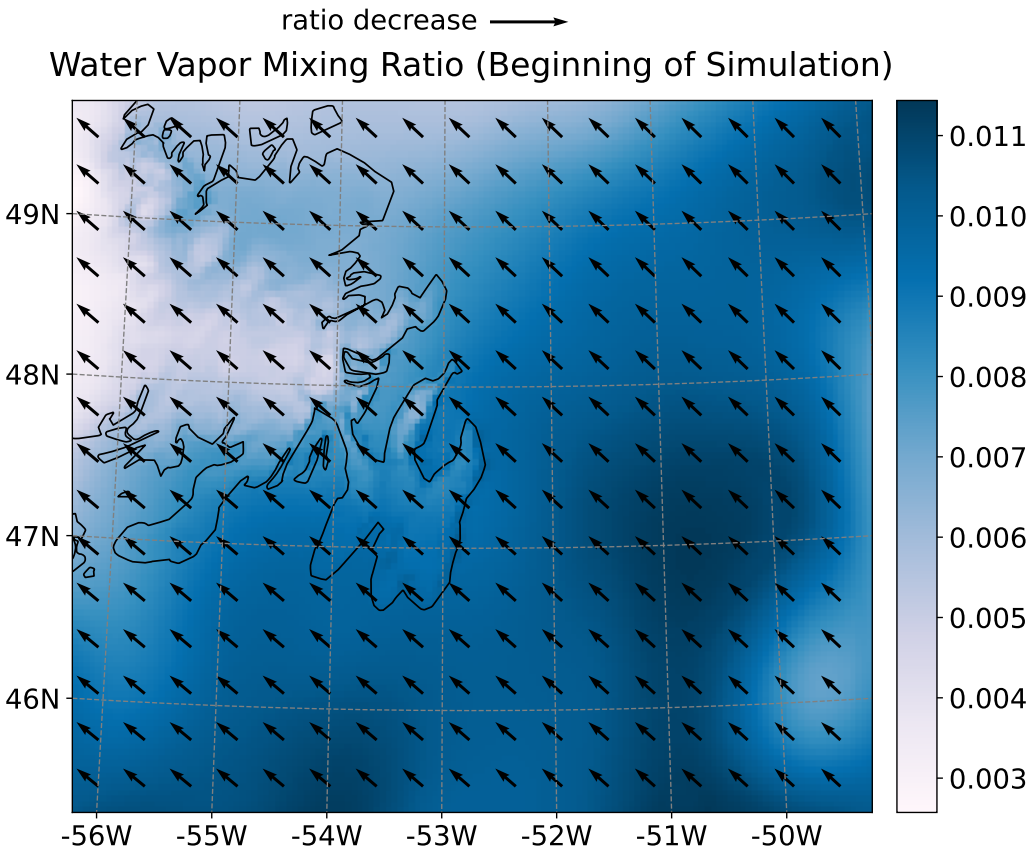

Figure 12: Water-vapor Mixing Ratio on Sep 4, 2019

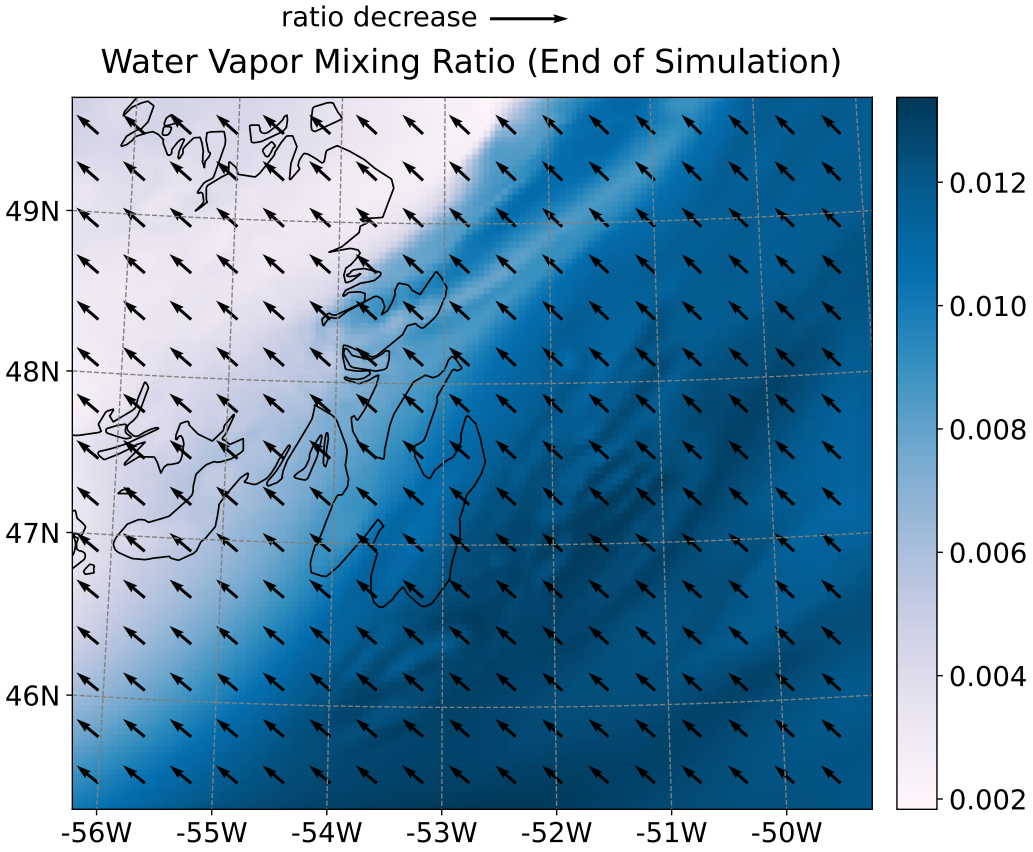

Figure 13: Water-vapor Mixing Ratio on Sep 6, 2019

3.2 Time Series Plots

3.2.1 Surface Temperature over Time

The graph of temperature versus time (Figure 14) shows a gradual increase in temperature in the first three quarters and a sudden decline in the last quarter of the simulation.

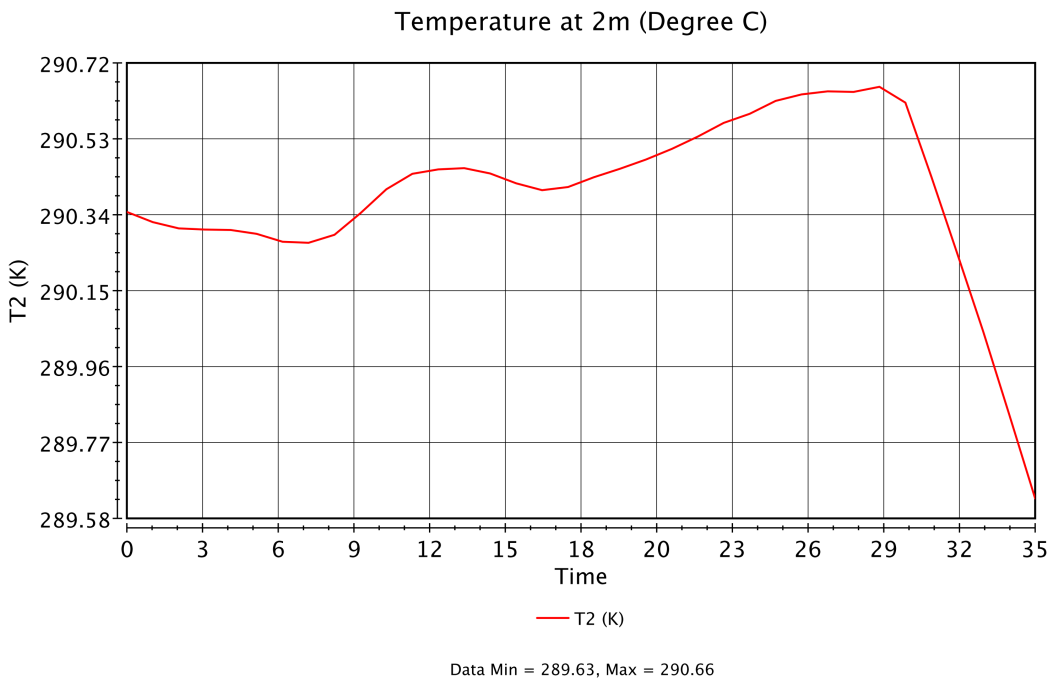

Figure 14: Surface Temperature over Time

3.2.2 Surface Pressure over Time

The graph of surface pressure versus time (Figure 15) shows a significant decrease for most of the simulation followed by a small increase in the end.

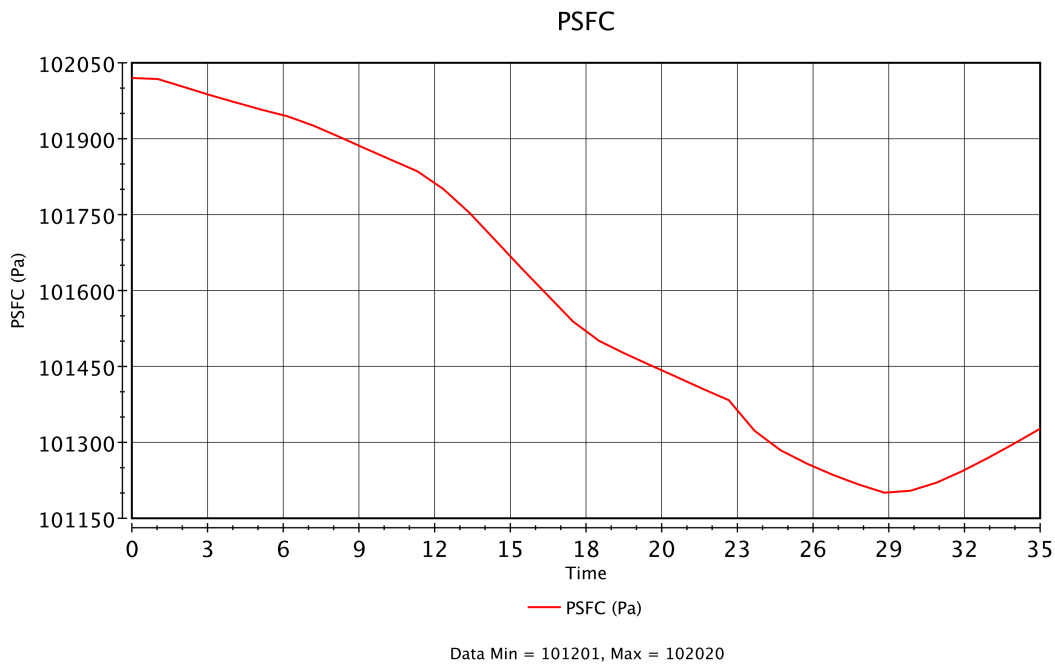

Figure 15: Surface Pressure over Time

3.2.3 Mid-height Pressure over Time

The graph of mid-height pressure versus time (Figure 16) is very similar to the graph of surface pressure versus time and shows a significant decrease for most of the simulation followed by a small increase in the end.

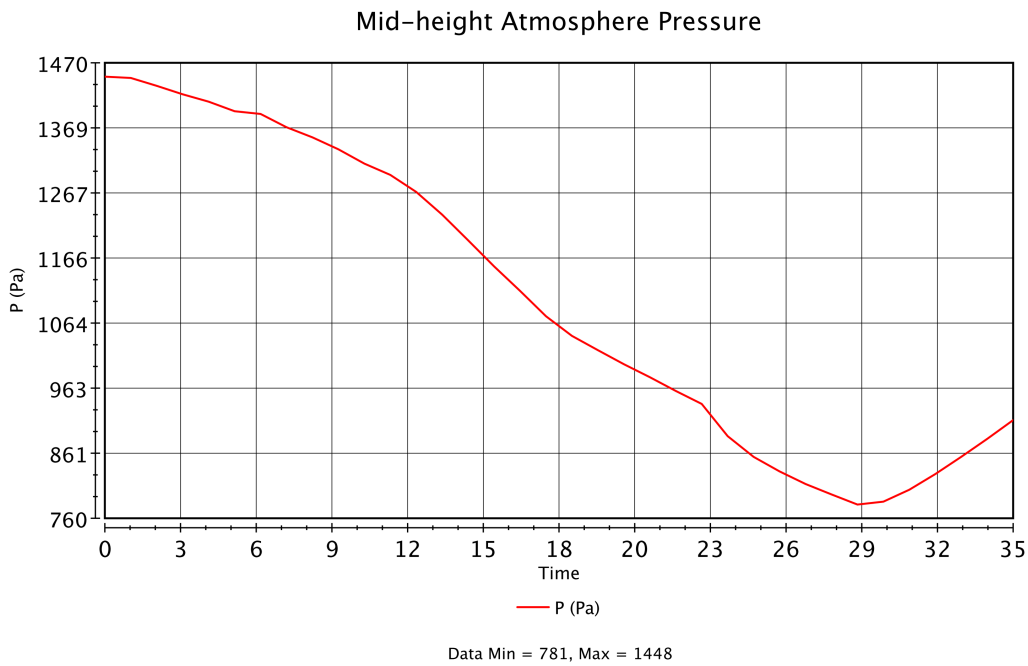

Figure 16: Mid-height Pressure over Time

3.2.4 Wind Velocity over Time

The graph of U versus time (Figure 17) shows a sudden decrease in the first quarter, a steep increase in the second and third quarters, and slow magnitude oscillations in the last quarter of the simulation. Graph of V versus time (Figure 18) shows overall changes with higher magnitudes. This graph conveys a gradual increase in the first three quarters of the simulation and a sudden decrease in the last quarter of the simulation.

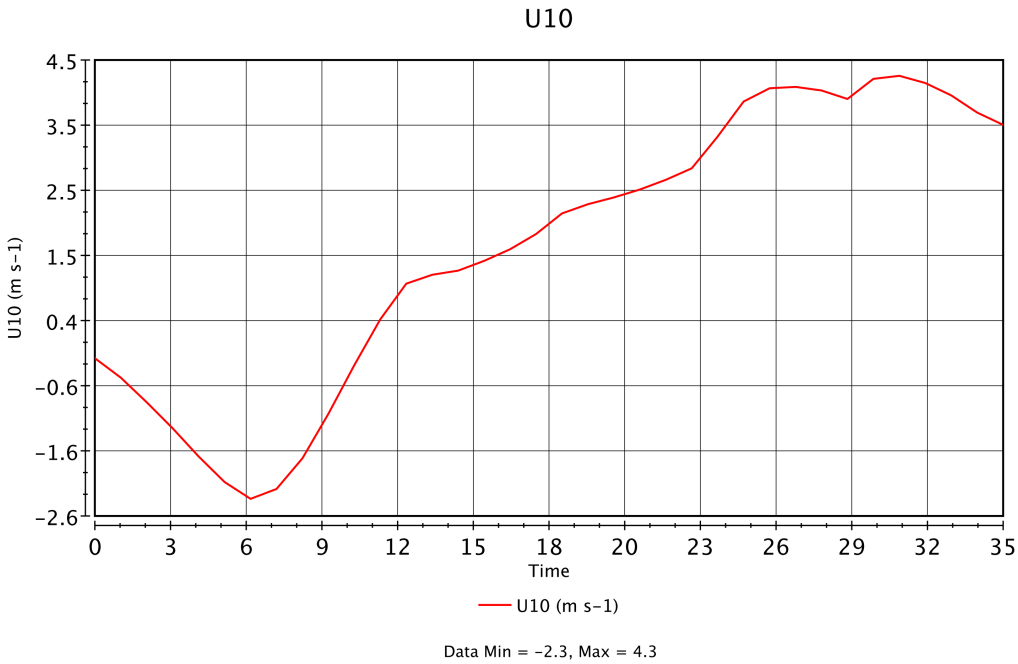

Figure 17: U (velocity in x-direction) over Time

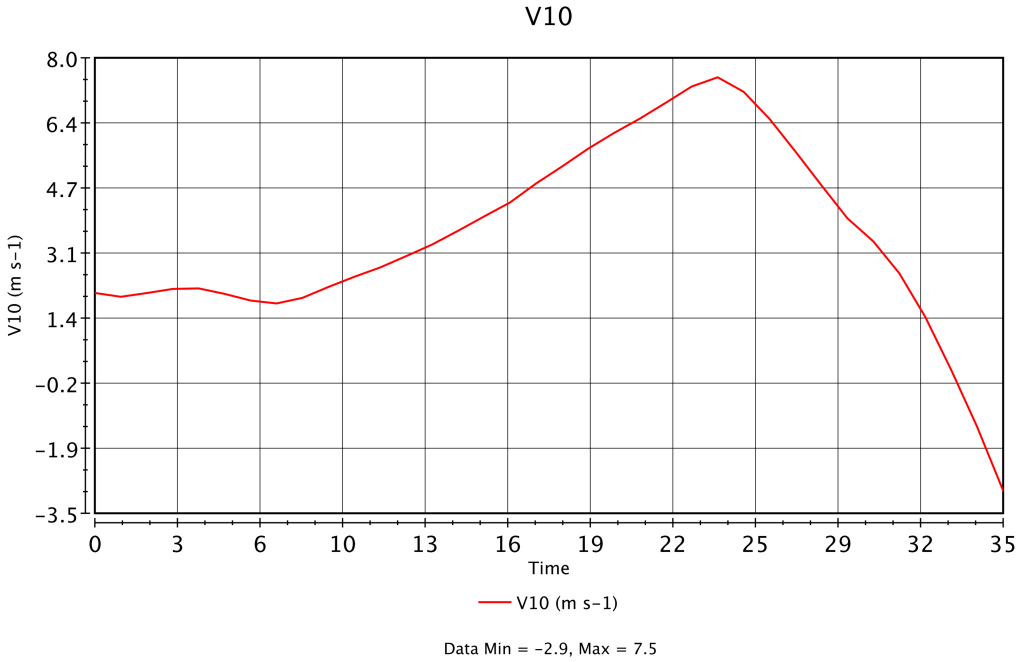

Figure 18: V (velocity in y-direction) over Time

4 Conclusion and Future Work

4.1 Wind, Temperature, and Pressure

In the results section, we explained that throughout the simulation, both surface pressure and mid-height pressure decrease. This change affects the shape of the equal pressure levels in the region. The change in the pressure levels can be observed in the shape of the contour lines of the surface and mid-height pressure graphs. Understandably, this change has a higher magnitude at a higher distance from the surface so it creates more obvious differences between the two mid-height pressure graphs. Curiously, the contour lines of the mid-height pressure are more ordered at the beginning and the end of the simulation. Observing these results, one might associate pressure level changes with local variables at lower heights and global factors at higher heights of the atmosphere.

In addition, It can also be observed that throughout the simulation as the lower-pressure air dominates the region under study, the temperature drops over the land and the water vapor ratio also decreases significantly. As a result of these changes, at the beginning of the simulation, the land surface area is warmer and the sea surface area is colder, but at the end, the sea surface area is warmer than the land area.

Finally, it can be concluded that the final results of the weather research and forecast model agree with the expected results based on our Atmospheric Physics knowledge. To enhance the accuracy of the model in the long run, further customization of the model's Physical settings based on the time and location of the simulation is necessary for a comprehensive future investigation of the WRF model.

References

[1] National Centre for Atmospheric Research. *User's Guide for Advanced Research WRF (ARW) Modeling System Version 2.2*. June 2007. URL: https://homepages.see.leeds.ac.uk/~lecrrb/wrf/aRWUsersGuide.pdf (visited on 04/16/2024).

[2] George D. Greenwade. "The Comprehensive Tex Archive Network (CTAN)". In: *TUGBoat* 14.3 (1993), pp. 342–351.

[3] W. C. Skamarock et al. "A Description of the Advanced Research WRF Version 3". In: (2008). DOI: 10.5065/D68S4MVH.

[4] University of Waterloo. *WRF Tutorial*. June 27, 2019. URL: https://wiki.math.uwaterloo.ca/fluidswiki/index.php?title=WRF_Tutorial (visited on 04/16/2024).

[5] Louis J. Wicker and William C. Skamarock. "Time-Splitting Methods for Elastic Models Using Forward Time Schemes". In: *Monthly Weather Review* 130.8 (2002), pp. 2088–2097. DOI: 10.1175/1520-0493(2002)130<2088:TSMFEM>2.0.CO;2. URL: https://journals.ametsoc.org/view/journals/mwre/130/8/1520-0493_2002_130_2088_tsmfem_2.0.co_2.xml.